



FAIR 3D HERITAGE

Documentation, Publication and Preservation
of the Digital 3D Heritage on the Web

From music rooms in the illustrated press to FAIR 3D heritage: web-based infrastructures, provenance, and preservation in Digital Art History

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Abstract

This poster presents an ongoing research project in Art History and Heritage Studies that investigates historic music rooms as multisensory and social environments, reactivated today through digital infrastructures, open data, and 3D reconstruction. Its central contribution is methodological and infrastructural: it implements and connects interoperable tools that allow heterogeneous research outputs (metadata, iconographic references, and 3D models) to circulate within a FAIR-aligned ecosystem, from historical source to museum display.

The project reinterprets early twentieth-century Portuguese domestic spaces associated with musical and artistic practices (Tavares & Barreira, 2023, 2024), drawing on the illustrated press, particularly *Ilustração Portuguesa* (1903–1924). These materials are processed and transformed into structured and interoperable data within a Wikibase Cloud instance (<https://music-rooms.wikibase.cloud>) developed by the first author, with the support of CODA/FLUP. The infrastructure, an online collaborative repository for art-historical and musicological data, implements persistent identifiers, version control, and semantic documentation (Lindemann et al., 2025; Barbosa et al., 2025; Trigo et al., in press), preserving both the digital assets and the workflows behind their creation. Rather than ingesting the periodical wholesale, the dataset derives from a curated selection of issues documenting music rooms, and is being enriched with a second periodical, *A Arte Musical* (1899–1915). The resulting metadata, linked to Wikidata (Wikimedia Foundation, 2003), guarantees provenance, multilingual interoperability, and reusability across domains and institutions.

Two strands of 3D work illustrate this workflow while reflecting the differing survival conditions of music rooms. For example, the music room of the Teixeira Lopes atelier, documented in *Ilustração Portuguesa*, survives largely intact: catalogued as structured data in Wikibase, it is recorded through a LiDAR and drone survey, enabling a 3D reconstruction of the room. By contrast, the music room of Alfredo Keil, documented in *Ilustração Portuguesa* but also in *A Arte Musical*, no longer exists; here the surviving instruments, now held at the Museu Nacional da Música (Mafra, Lisbon district), are digitised in 3D, the periodical descriptions providing context for their original setting. Across both cases the FAIR principles (Wilkinson et al., 2016) are operationalised concretely: sources, instruments, and models receive persistent identifiers (Findable); data, paradata, and outputs are deposited in openly accessible repositories with documented licensing (Accessible); standardised vocabularies and Wikidata alignment ensure semantic interoperability (Interoperable); and versioned, documented workflows allow reconstructions to be reused, verified, and updated (Reusable).

Here, repository design integrates preservation strategy, publication, and intellectual accountability, with infrastructure functioning as research environment, publication platform, and preservation system. The project then extends into the Museu Nacional da Música, where these reconstructions, semantic data, and related visualisations inform the museum's digital mediation strategy, allowing assets to move between research and exhibition platforms while retaining provenance and metadata integrity.

Integrating digital humanities methods with FAIR data and open science frameworks (European Commission, 2019), the research positions the reconstruction of historic music rooms as a case study in sustainable 3D heritage practice. It argues that digital preservation infrastructures must function both as technical systems and as frameworks of governance, combining semantic precision, ethical curation, and institutional responsibility, and that the value of 3D heritage depends on its integration within transparent, reusable data environments rather than on the models alone.

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